

Library Management System

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Console-Based Database Management Application

Purpose:

This project was developed to demonstrate the implementation of a simple library management system using Python and MySQL. The application supports administrator operations such as user management, book management, and search functionality.

A. Administrator Login

This screen demonstrates secure administrator authentication before accessing management features.

Login options are displayed

```
=== Library Management System ===
1. Admin Login
2. User Login
3. Exit
Enter choice: |
```

Administrator chooses 'Admin Login' and enters Username and Password.

```
=== Library Management System ===
1. Admin Login
2. User Login
3. Exit
Enter choice: 1
Enter admin credentials:
Username: Admin
Password:
Admin login successful!
```

Functionalities Demonstrated

- Username verification
- Password authentication
- Protected password entry using 'getpass' module of Python
- Role-based system access

B. Display Admin Menu

Post login the menu options for the Administrator are displayed

```
Admin login successful!

--- Admin Menu ---
1. Add User
2. Delete User
3. Update User
4. Search User
5. Display All Users
6. Add Book
7. Delete Book
8. Update Book
9. Search Book
10. Display All Books
11. Issue Book
12. Return Book
13. Logout
Enter your choice: |
```

Functionalities Demonstrated

- Database retrieval
- Structured Result display

C. Display All Users

The administrator can view all registered users stored in the database.

```
--- Admin Menu ---
1. Add User
2. Delete User
3. Update User
4. Search User
5. Display All Users
6. Add Book
7. Delete Book
8. Update Book
9. Search Book
10. Display All Books
11. Issue Book
12. Return Book
13. Logout
Enter your choice: 5

List of Users:
ID: 1 Name: Alice Email: alice@example.com Phone: 1234567890
ID: 3 Name: Ryan Dasgupta Email: tunirdasgupta08@gmail.com Phone: 8697418805
```

Functionalities Demonstrated

- Database retrieval
- Structured Result display

D. Add New User

This feature allows the administrator to register a new user into the system.

```
--- Admin Menu ---
1. Add User
2. Delete User
3. Update User
4. Search User
5. Display All Users
6. Add Book
7. Delete Book
8. Update Book
9. Search Book
10. Display All Books
11. Issue Book
12. Return Book
13. Logout
Enter your choice: 1

Add New User
Enter Name: Roger Federer
Enter Email: roger.federer@gmail.com
Enter Phone: 765544789
Enter Password: wimbledon@123
User added successfully with ID 4
```

Library Management System

Now, administrator can check that the user was actually added using the 'Display All Users' feature.

```
--- Admin Menu ---
1. Add User
2. Delete User
3. Update User
4. Search User
5. Display All Users
6. Add Book
7. Delete Book
8. Update Book
9. Search Book
10. Display All Books
11. Issue Book
12. Return Book
13. Logout
Enter your choice: 5

List of Users:
ID: 1 Name: Alice Email: alice@example.com Phone: 1234567890
ID: 3 Name: Ryan Dasgupta Email: tunirdasgupta08@gmail.com Phone: 8697418805
ID: 4 Name: Roger Federer Email: roger.federer@gmail.com Phone: 765544789
```

Functionalities Demonstrated

- User input handling
- Database insertion

E. Delete User

First, display all users in the system using the 'Display All Users' feature.

```
--- Admin Menu ---
1. Add User
2. Delete User
3. Update User
4. Search User
5. Display All Users
6. Add Book
7. Delete Book
8. Update Book
9. Search Book
10. Display All Books
11. Issue Book
12. Return Book
13. Logout
Enter your choice: 5

List of Users:
ID: 1 Name: Alice Email: alice@example.com Phone: 1234567890
ID: 3 Name: Ryan Dasgupta Email: tunirdasgupta08@gmail.com Phone: 8697418805
ID: 4 Name: Roger Federer Email: roger.federer@gmail.com Phone: 765544789
```

The 'Delete User' feature allows the administrator to delete an user from the system.

```
--- Admin Menu ---
1. Add User
2. Delete User
3. Update User
4. Search User
5. Display All Users
6. Add Book
7. Delete Book
8. Update Book
9. Search Book
10. Display All Books
11. Issue Book
12. Return Book
13. Logout
Enter your choice: 2
Enter User ID to delete: 4
Are you sure you want to delete user 4? (y/n): y
User deleted successfully.
```

Library Management System

Now, the administrator can check that the user was actually deleted using the 'Display All Users' feature.

```
--- Admin Menu ---
1. Add User
2. Delete User
3. Update User
4. Search User
5. Display All Users
6. Add Book
7. Delete Book
8. Update Book
9. Search Book
10. Display All Books
11. Issue Book
12. Return Book
13. Logout
Enter your choice: 5

List of Users:
ID: 1 Name: Alice Email: alice@example.com Phone: 1234567890
ID: 3 Name: Ryan Dasgupta Email: tunirdasgupta08@gmail.com Phone: 8697418805
```

Functionalities Demonstrated

- User input handling
- Database deletion

F. Display All Books

The administrator can look up all books stored in the database using the 'Display All Users' feature.

```
--- Admin Menu ---
1. Add User
2. Delete User
3. Update User
4. Search User
5. Display All Users
6. Add Book
7. Delete Book
8. Update Book
9. Search Book
10. Display All Books
11. Issue Book
12. Return Book
13. Logout
Enter your choice: 10

List of Books:
ID: 1 Title: The Great Gatsby Author: F. Scott Fitzgerald Year: 1984 Copies: 3
ID: 2 Title: Nineteen Eighty-four Author: George Orwell Year: 1949 Copies: 2
ID: 3 Title: The Old Man and the Sea Author: Ernst Hemingway Year: 1952 Copies: 3
```

Functionalities Demonstrated

- Database retrieval
- Structured Result display

G. Add New Book

The administrator can add new books to the library inventory.

```
--- Admin Menu ---
1. Add User
2. Delete User
3. Update User
4. Search User
5. Display All Users
6. Add Book
7. Delete Book
8. Update Book
9. Search Book
10. Display All Books
11. Issue Book
12. Return Book
13. Logout
Enter your choice: 6

Add New Book
Title: Sonar Kella
Author: Satyajit Ray
Publisher: Ananda Publishers
Publication Year: 1965
Number of copies: 4
Book added successfully with ID 4
```

Adminstrator can check that the book was actually added using the 'Display All Books feature.

```
--- Admin Menu ---
1. Add User
2. Delete User
3. Update User
4. Search User
5. Display All Users
6. Add Book
7. Delete Book
8. Update Book
9. Search Book
10. Display All Books
11. Issue Book
12. Return Book
13. Logout
Enter your choice: 10

List of Books:
ID: 1 Title: The Great Gatsby Author: F. Scott Fitzgerald Year: 1984 Copies: 3
ID: 2 Title: Nineteen Eighty-four Author: George Orwell Year: 1949 Copies: 2
ID: 3 Title: The Old Man and the Sea Author: Ernst Hemingway Year: 1952 Copies: 3
ID: 4 Title: Sonar Kella Author: Satyajit Ray Year: 1965 Copies: 4
```

Functionalities Demonstrated

- User input handling
- Database insertion

H. Search Book

The search functionality enables the administrator to locate books using keywords or titles. Partial search is allowed.

```
--- Admin Menu ---
1. Add User
2. Delete User
3. Update User
4. Search User
5. Display All Users
6. Add Book
7. Delete Book
8. Update Book
9. Search Book
10. Display All Books
11. Issue Book
12. Return Book
13. Logout
Enter your choice: 9
Enter book title or ID to search: Sonar
ID: 4 Title: Sonar Kella Author: Satyajit Ray Year: 1965 Copies: 4
```

Functionalities Demonstrated

- Search query handling
- Partial search
- Structured Result display

I. Update Book

Administrator displays all books in the system using the 'Display All Books' feature.

```
--- Admin Menu ---
1. Add User
2. Delete User
3. Update User
4. Search User
5. Display All Users
6. Add Book
7. Delete Book
8. Update Book
9. Search Book
10. Display All Books
11. Issue Book
12. Return Book
13. Logout
Enter your choice: 10

List of Books:
ID: 1 Title: The Great Gatsby Author: F. Scott Fitzgerald Year: 1984 Copies: 3
ID: 2 Title: Nineteen Eighty-four Author: George Orwell Year: 1949 Copies: 2
ID: 3 Title: The Old Man and the Sea Author: Ernst Hemingway Year: 1952 Copies: 3
ID: 4 Title: Sonar Kella Author: Satyajit Ray Year: 1965 Copies: 4
```

Library Management System

The 'Update Book' feature allows the administrator to update details of a book in the system. In this instance updating the Year to 1971 from 1965.

```
--- Admin Menu ---
1. Add User
2. Delete User
3. Update User
4. Search User
5. Display All Users
6. Add Book
7. Delete Book
8. Update Book
9. Search Book
10. Display All Books
11. Issue Book
12. Return Book
13. Logout
Enter your choice: 8
Enter Book ID to update: 4
Enter new data (leave blank to keep unchanged):
New Title:
New Author:
New Publisher:
New Year: 1971
New Copies:
Book updated successfully.
```

Administrator can display the updated information using the 'Display all Books' feature.

```
--- Admin Menu ---
1. Add User
2. Delete User
3. Update User
4. Search User
5. Display All Users
6. Add Book
7. Delete Book
8. Update Book
9. Search Book
10. Display All Books
11. Issue Book
12. Return Book
13. Logout
Enter your choice: 10

List of Books:
ID: 1 Title: The Great Gatsby Author: F. Scott Fitzgerald Year: 1984 Copies: 3
ID: 2 Title: Nineteen Eighty-four Author: George Orwell Year: 1949 Copies: 2
ID: 3 Title: The Old Man and the Sea Author: Ernst Hemingway Year: 1952 Copies: 3
ID: 4 Title: Sonar Kella Author: Satyajit Ray Year: 1971 Copies: 4
```

Functionalities Demonstrated

- User input handling
- Database update